

Design proposal for Mach Zander Interferometer
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Introduction

Mach zander interferometer can be used to generate an interference pattern to study phase shift difference of two beams split from the same source using a splitter or Y branch. In this proposal we use a Y Branch splitter. Mach Zander Interferometers have applications in Qunatum Entanglement, opto-electronic modulators, fiber optic communications, holography etc.

Theory

A basic unbalanced MZI consists is a two port device which has beam splitter/Y Branch followed by unidentical/asymmetric waveguides which produce the phase shift followed by beam/YBranch combiner. For an interferometer using a YBranch splitter and combiner the intensity is given by:

$$I_o = \frac{I_i}{2} [1 + \cos(\beta\Delta L)]$$

Where ΔL is the difference in path length and β is the propagation constant. The transfer function of the MZI is given by:

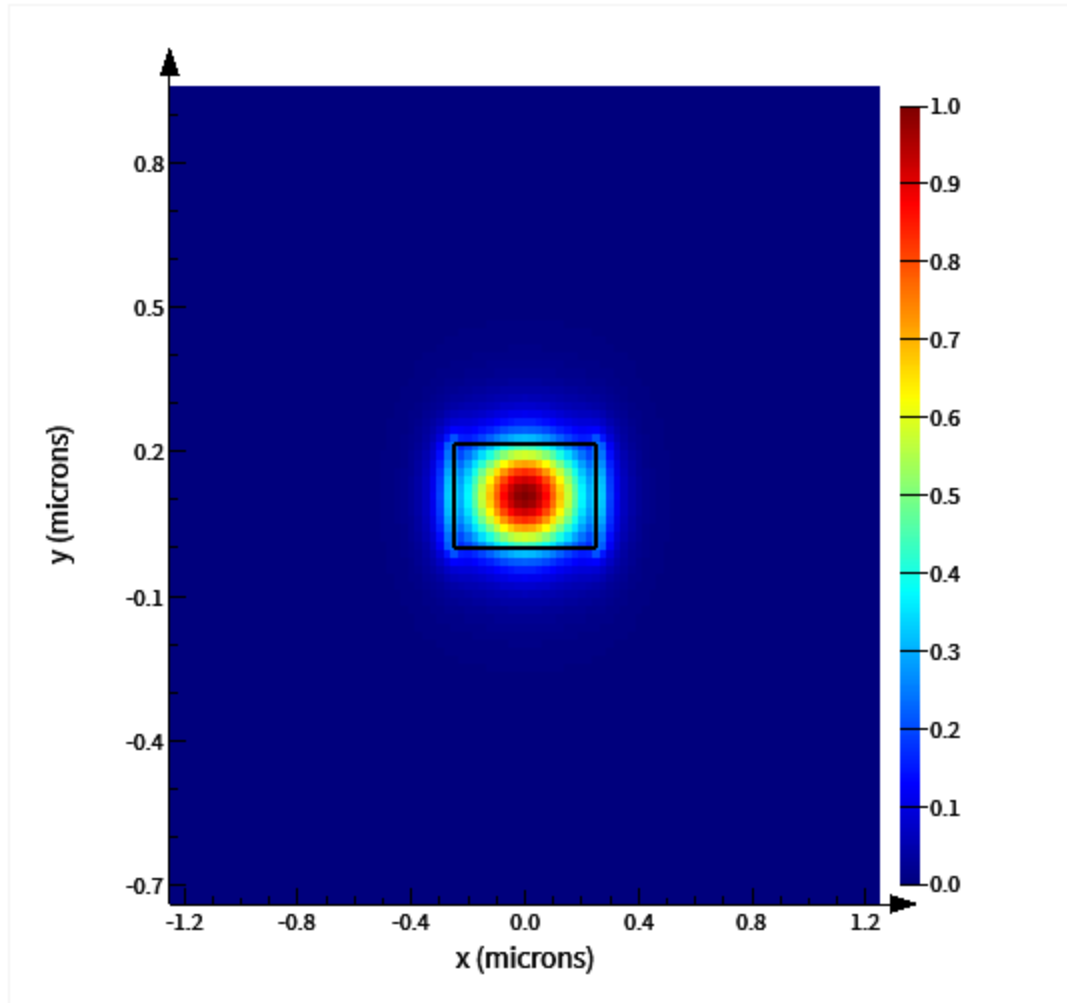
$$T = \frac{1}{2} [1 + \cos(\beta\Delta L)]$$

And the free spectral range is given by:

$$\text{FSR} = \frac{\lambda^2}{n_g \Delta L}$$

Modelling and Simulation:

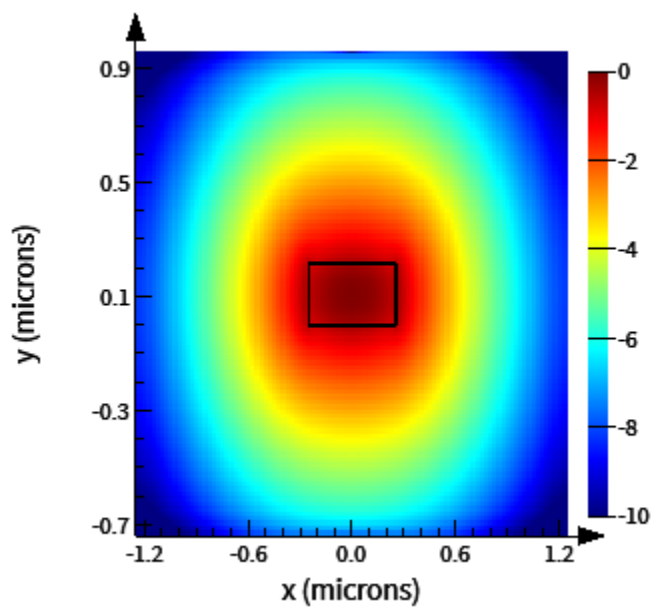
We choose 5 different path lengths for 500nm by 220nm waveguide (TE mode) and compare expected FSR with simulation for TE .



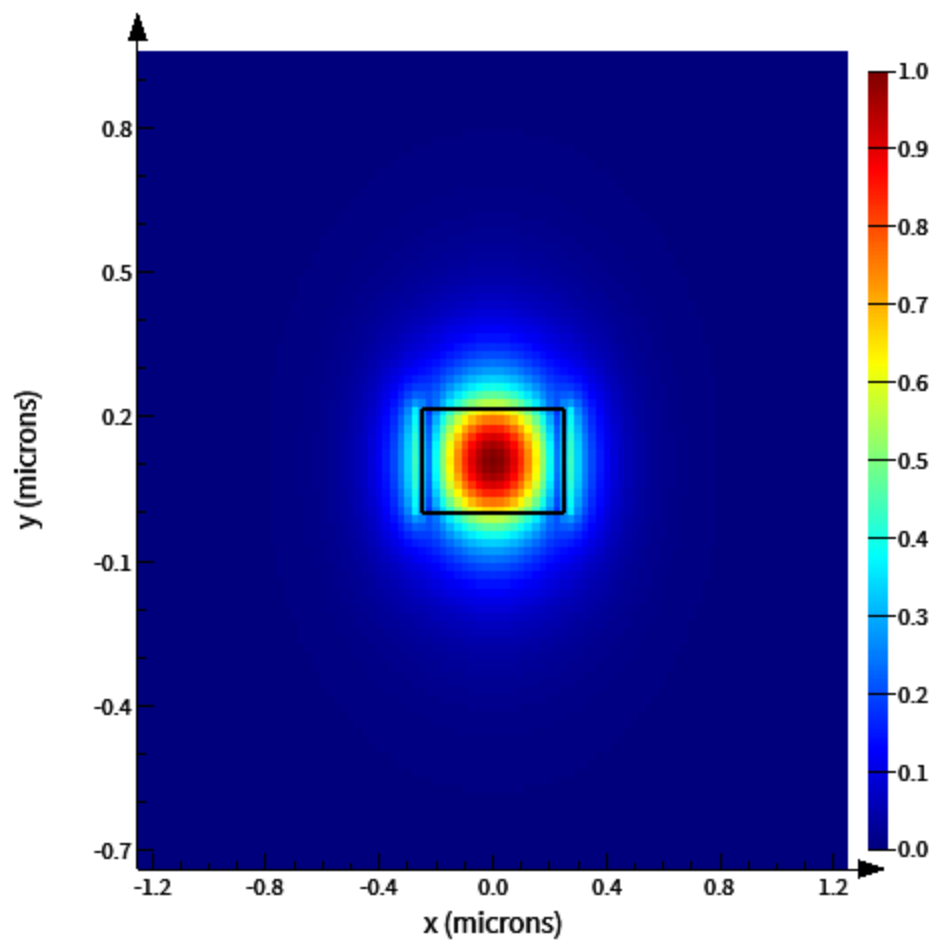
Electric field intensity of TE mode

mode #	effective index	wavelength (μm)	loss (dB/cm)	group index	T
1	2.503380+1.297784e-09i	1.5	0.00047218	4.196678+2.802684e-09i	99
2	1.836526+9.718270e-10i	1.5	0.00035358	4.007656+4.785884e-09i	4
3	1.532900+6.249788e-10i	1.5	0.00022739	2.815674+5.056738e-09i	70
4	1.366036+3.363229e-10i	1.5	0.00012237	2.098597+2.558310e-09i	8

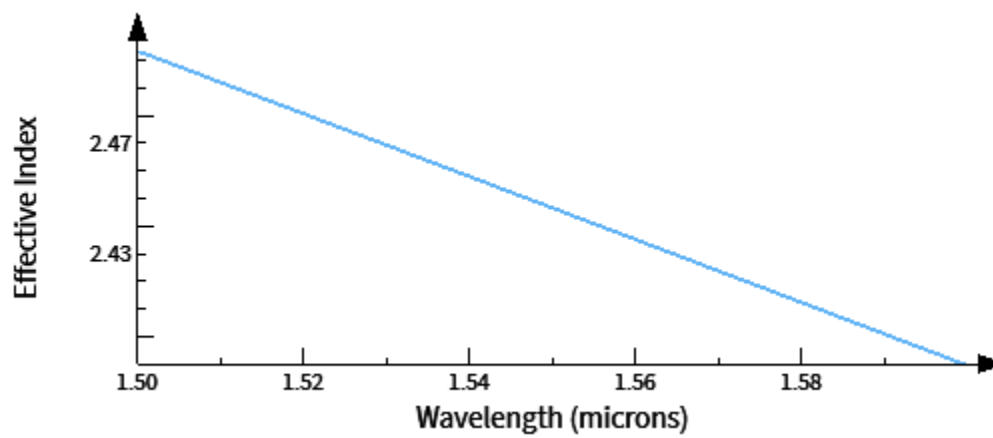
Effective indices



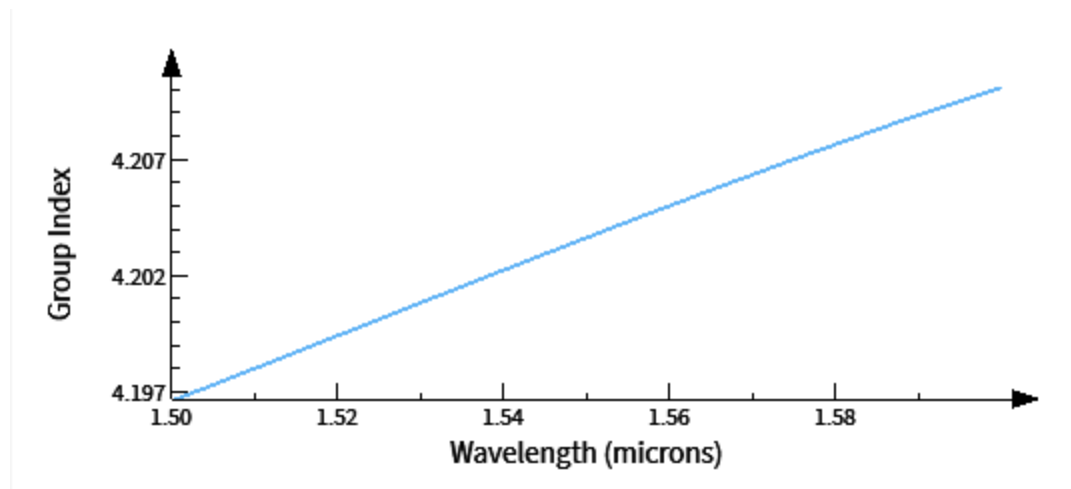
Electric field on log scale



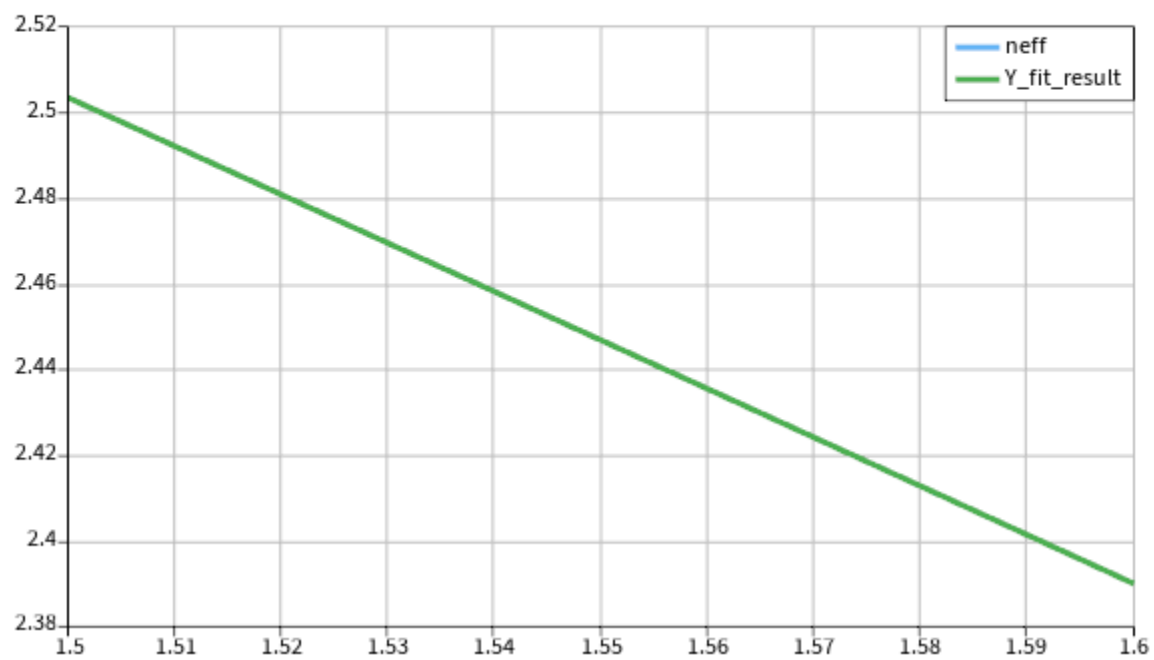
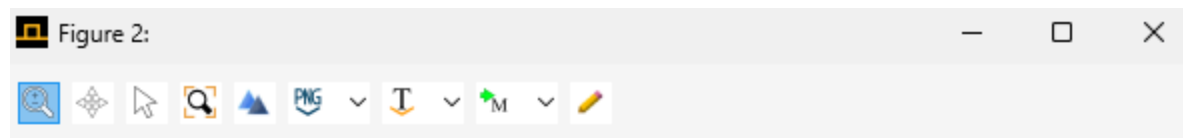
Ex



Effective index vs wavelength

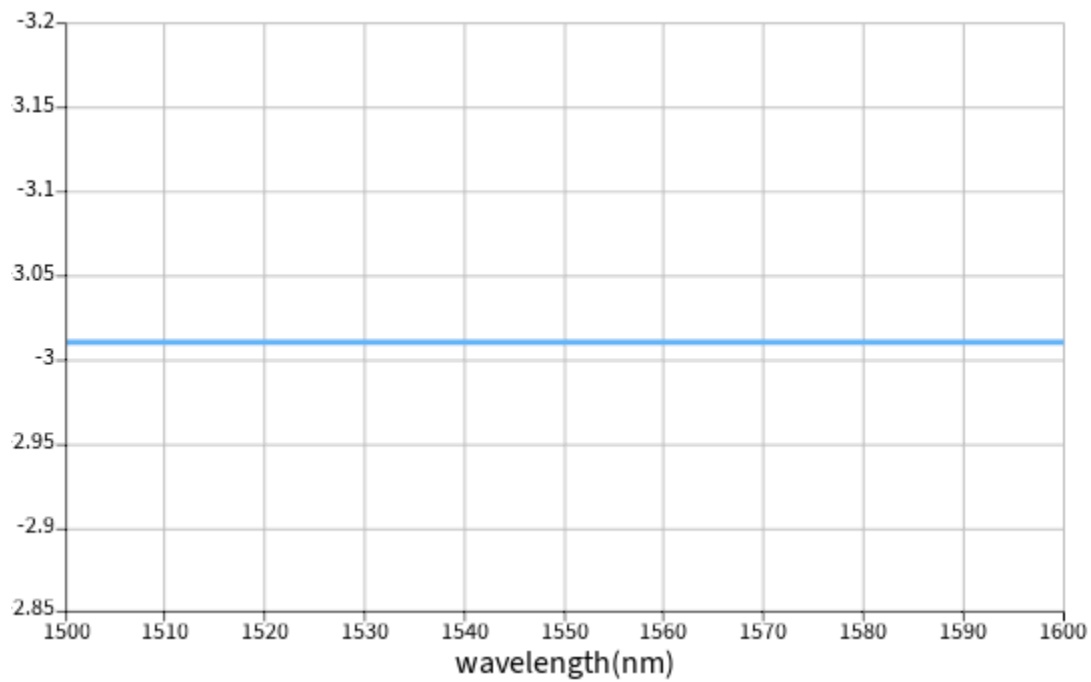


Group index vs wavelength

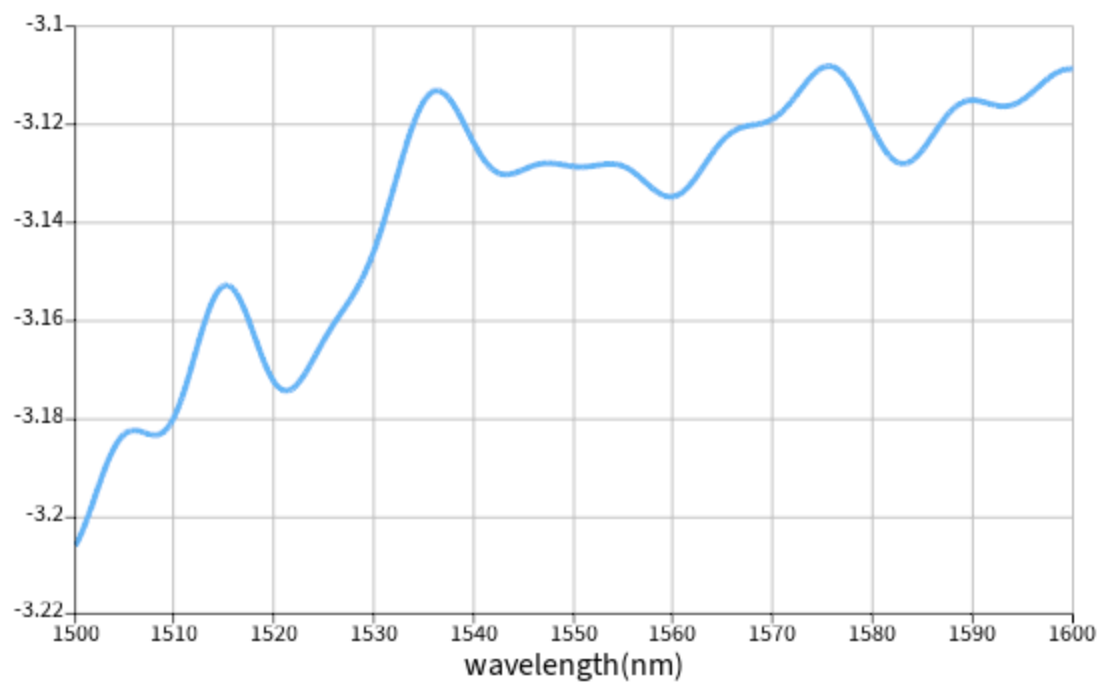


$$2.44682 - 1.13339 \cdot (\lambda - 1.55) - 0.0439366 \cdot (\lambda - 1.55)^2$$

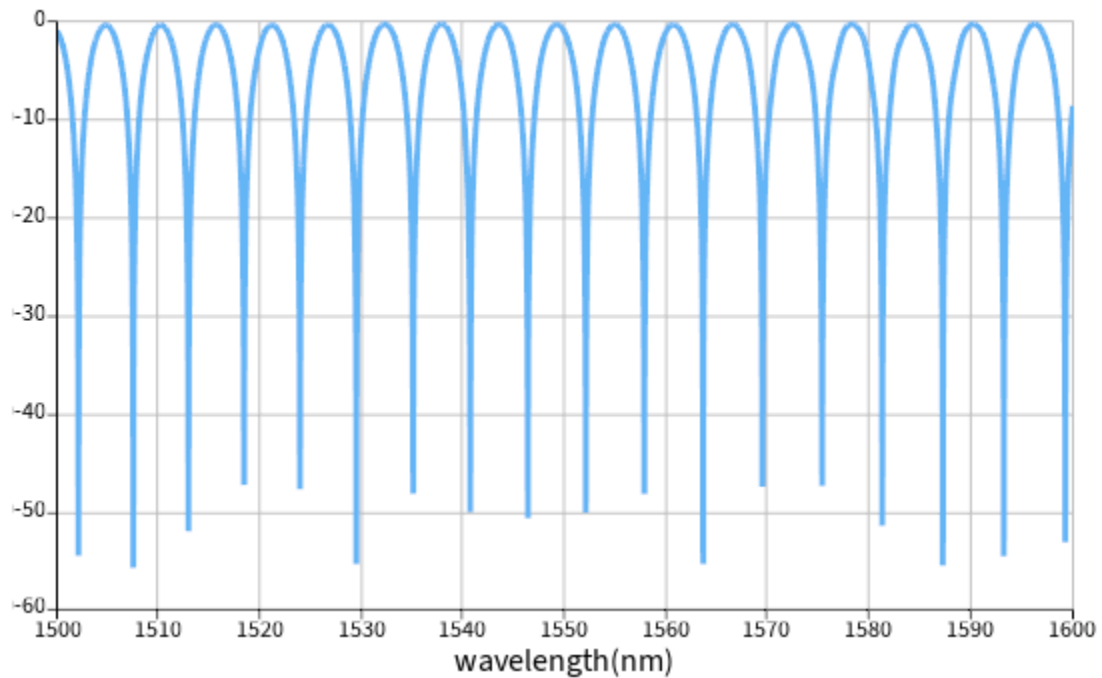
Compact model for waveguide calculated using Lumerical script



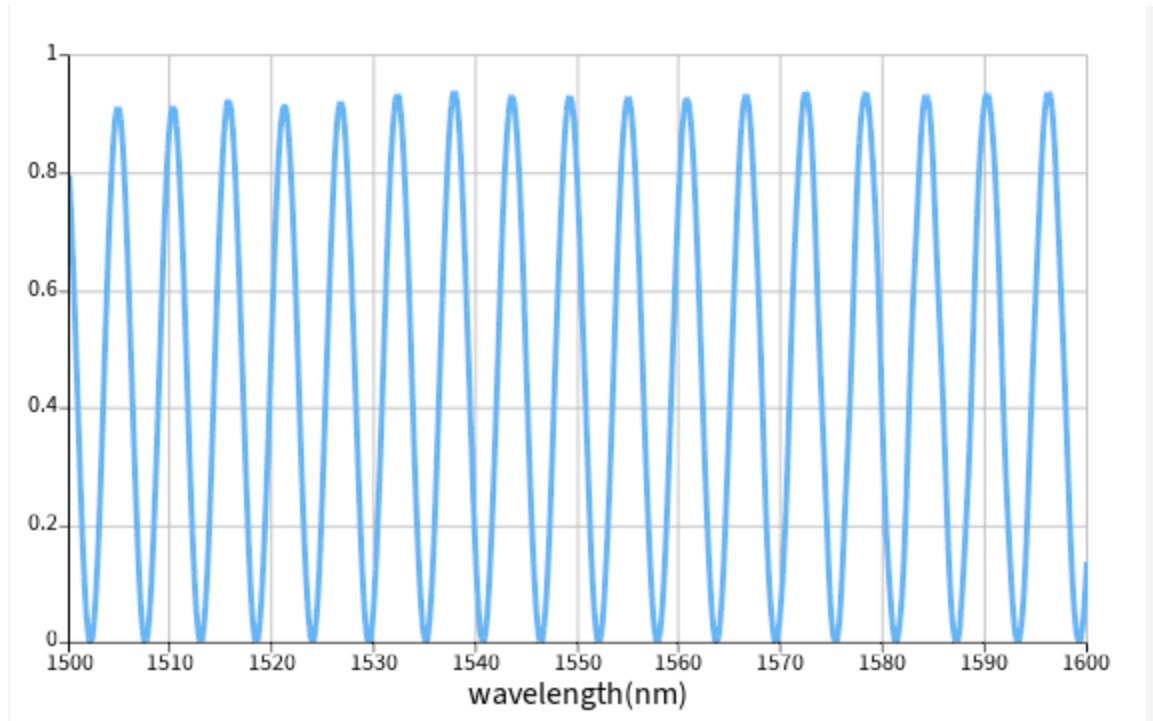
Insertion loss ideal y branch



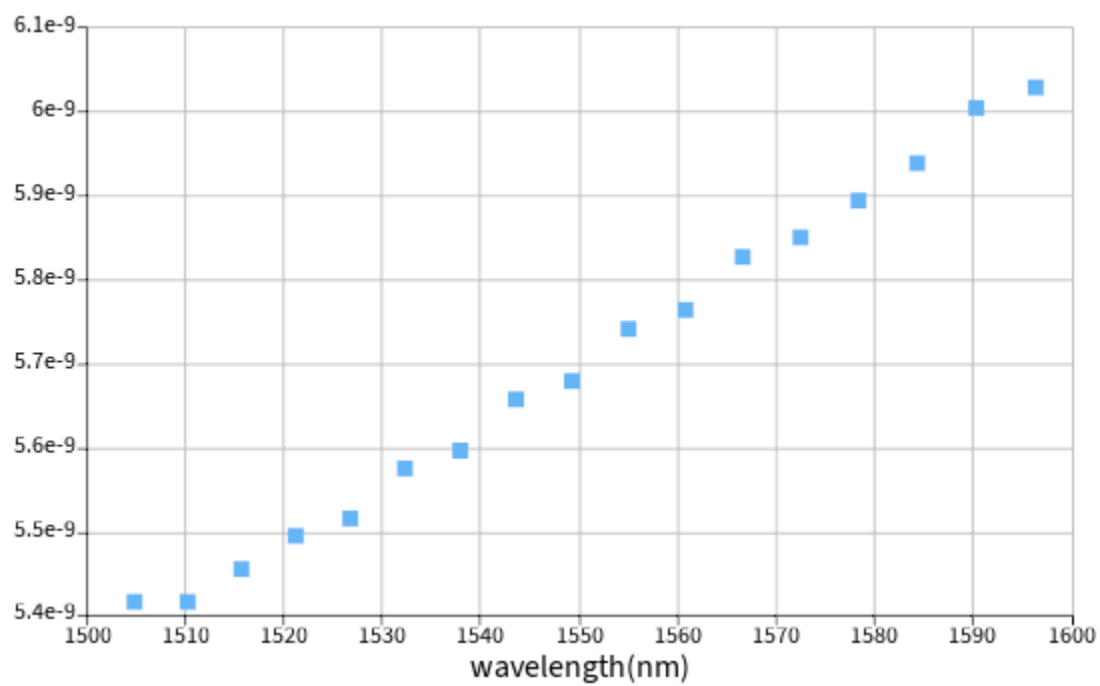
Insertion loss s parameter y branch



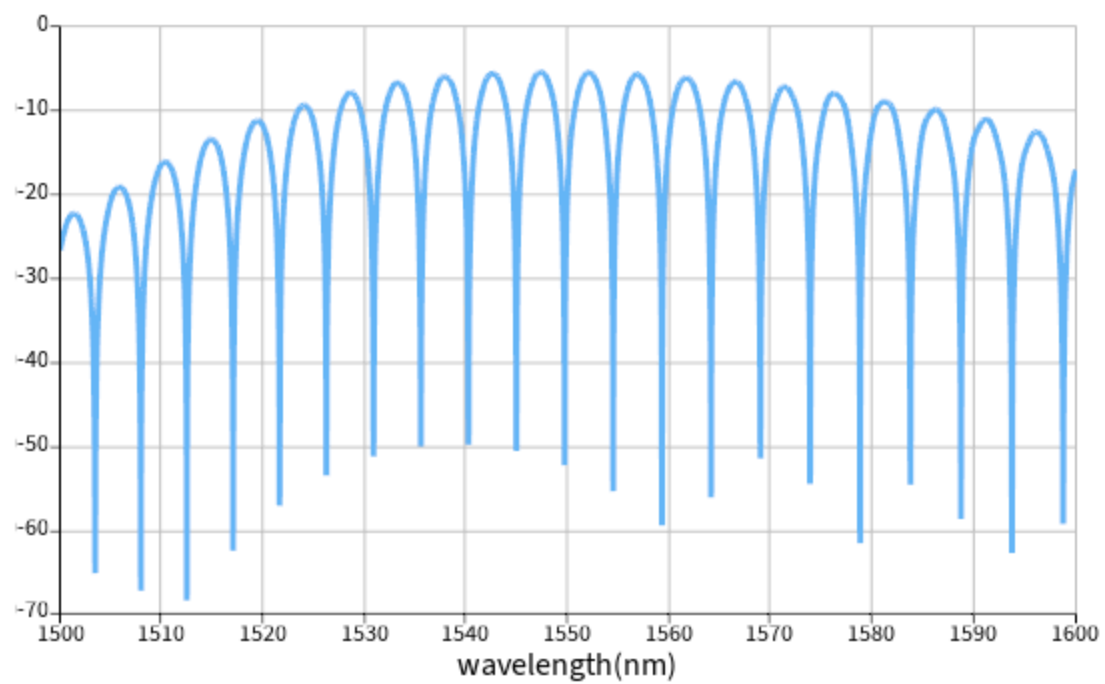
Spectrum of MZI delta L 100



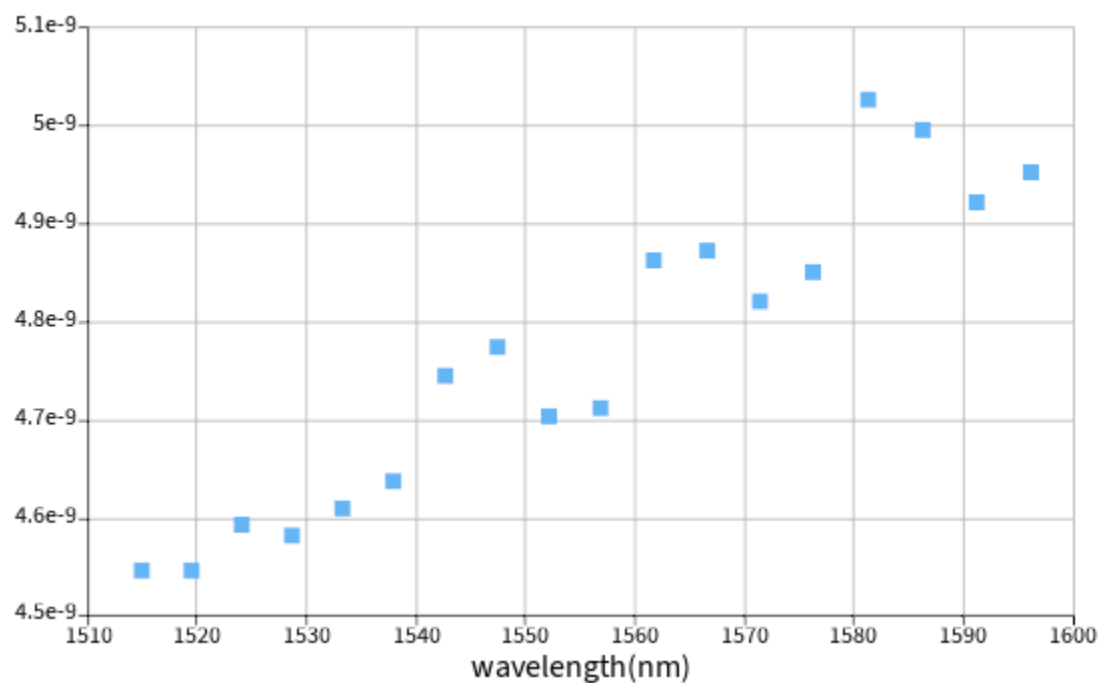
TE transmission delta L 100



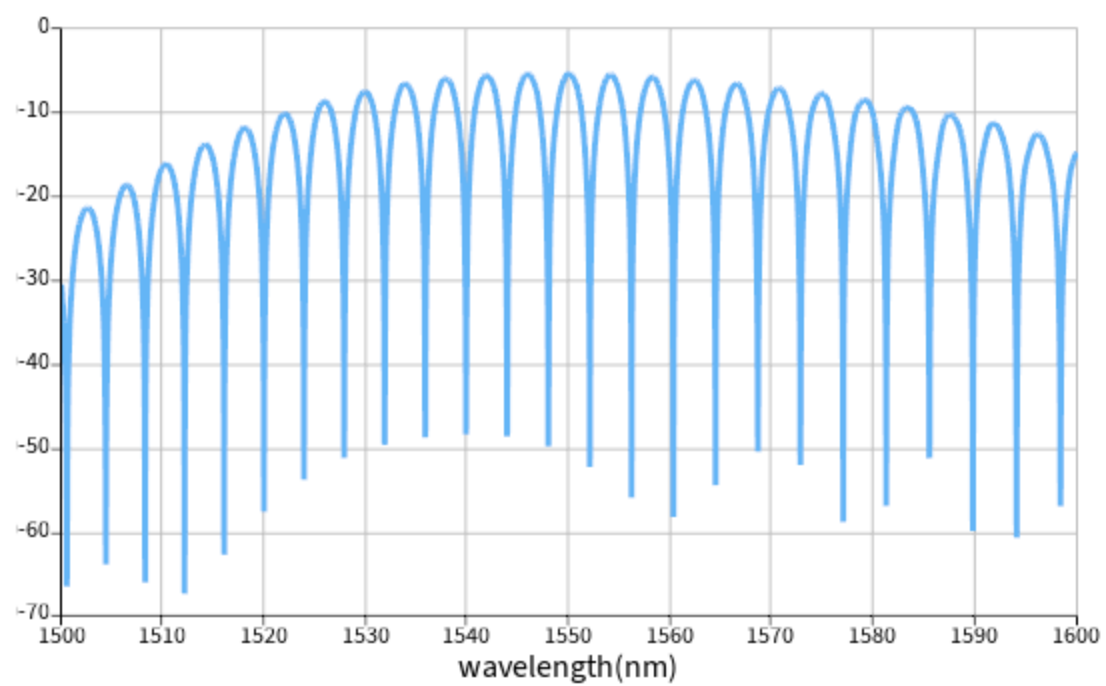
FSR delta L 100 expected FSR = 5.72024e-09



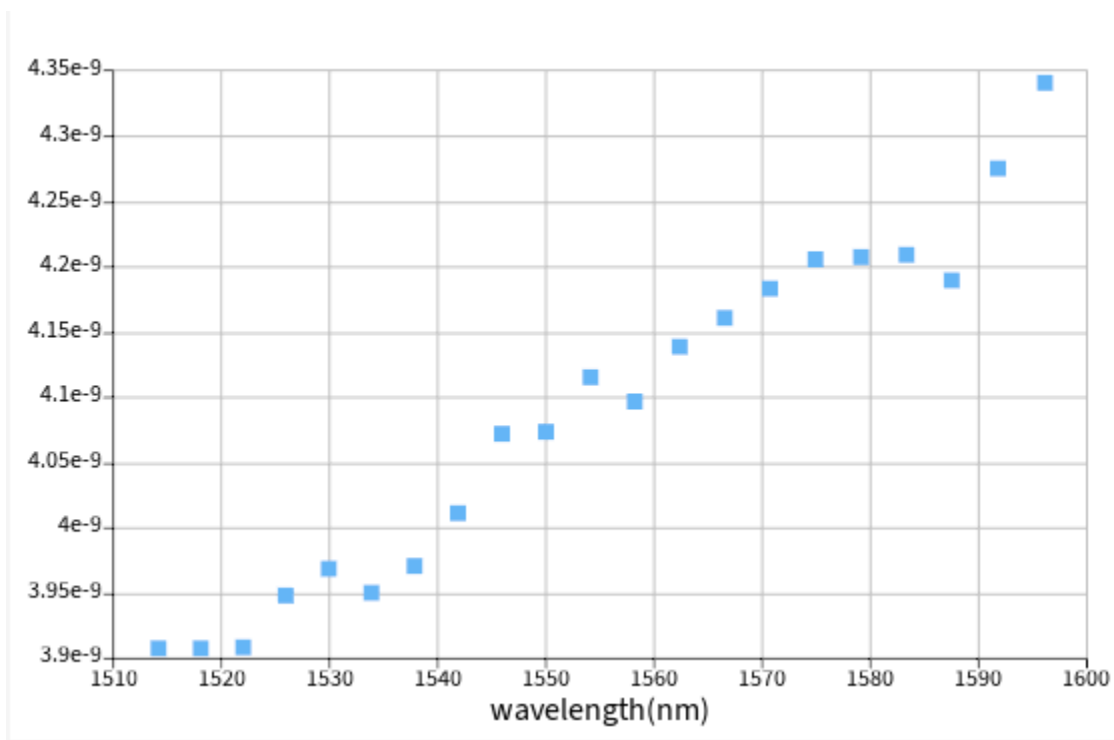
Spectrum of MZI delta L 120



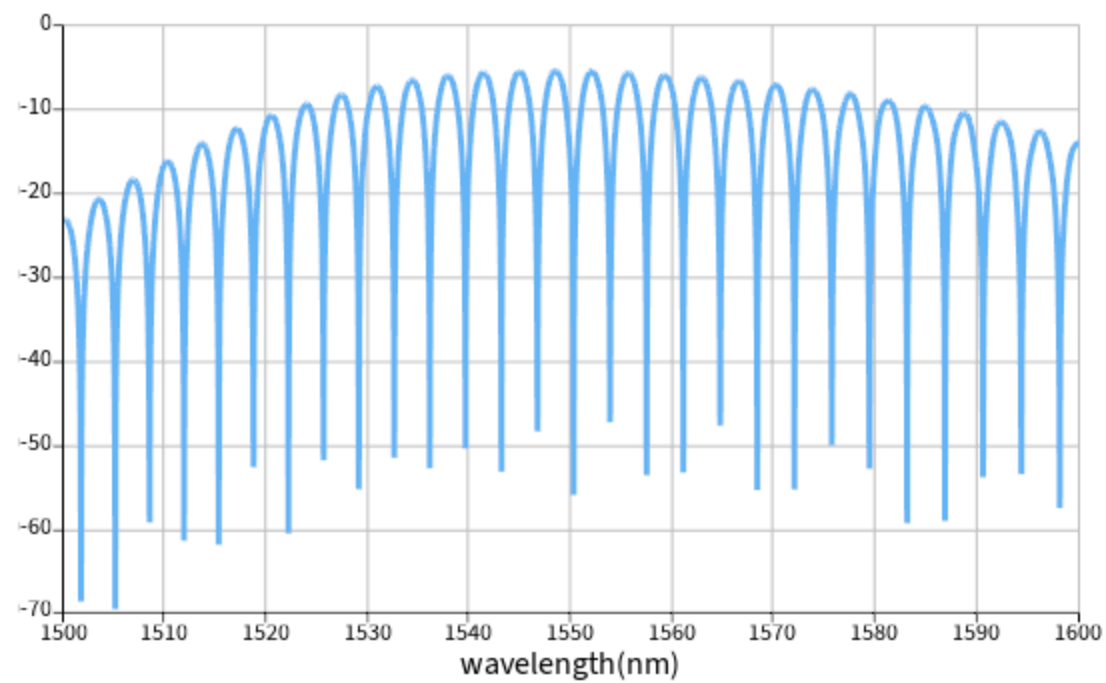
FSR delta L 120 expected FSR = 4.76687×10^{-9}



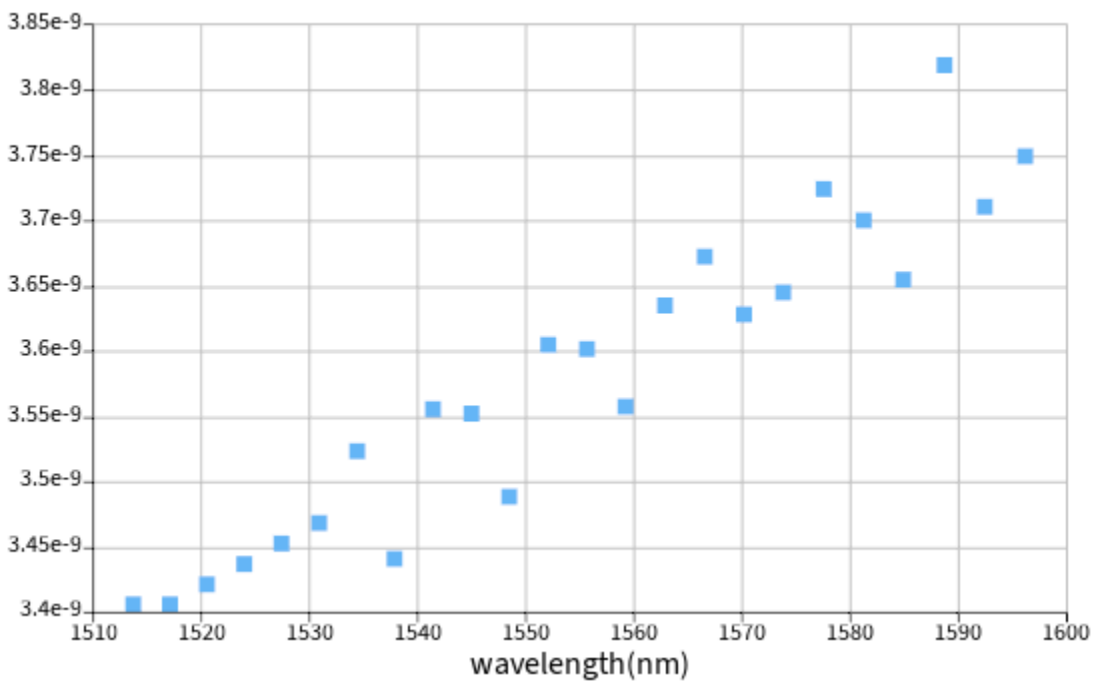
Spectrum of MZI delta L 140



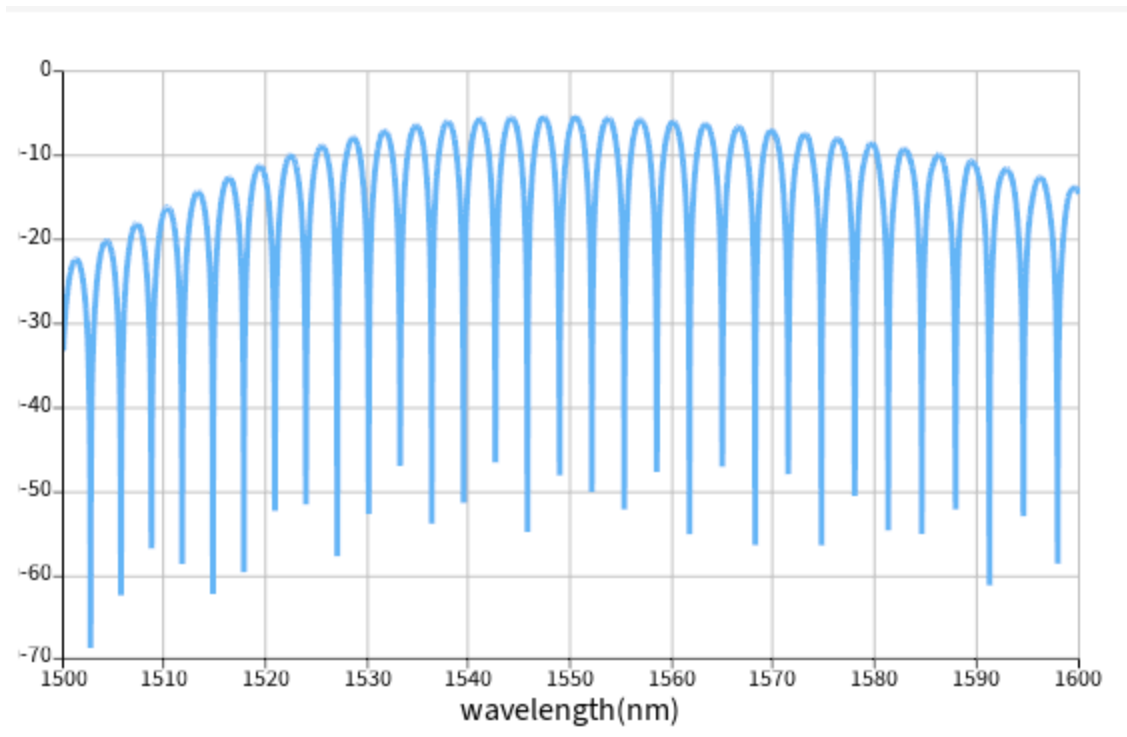
FSR delta L 140 expected FSR = 4.08588×10^{-9}



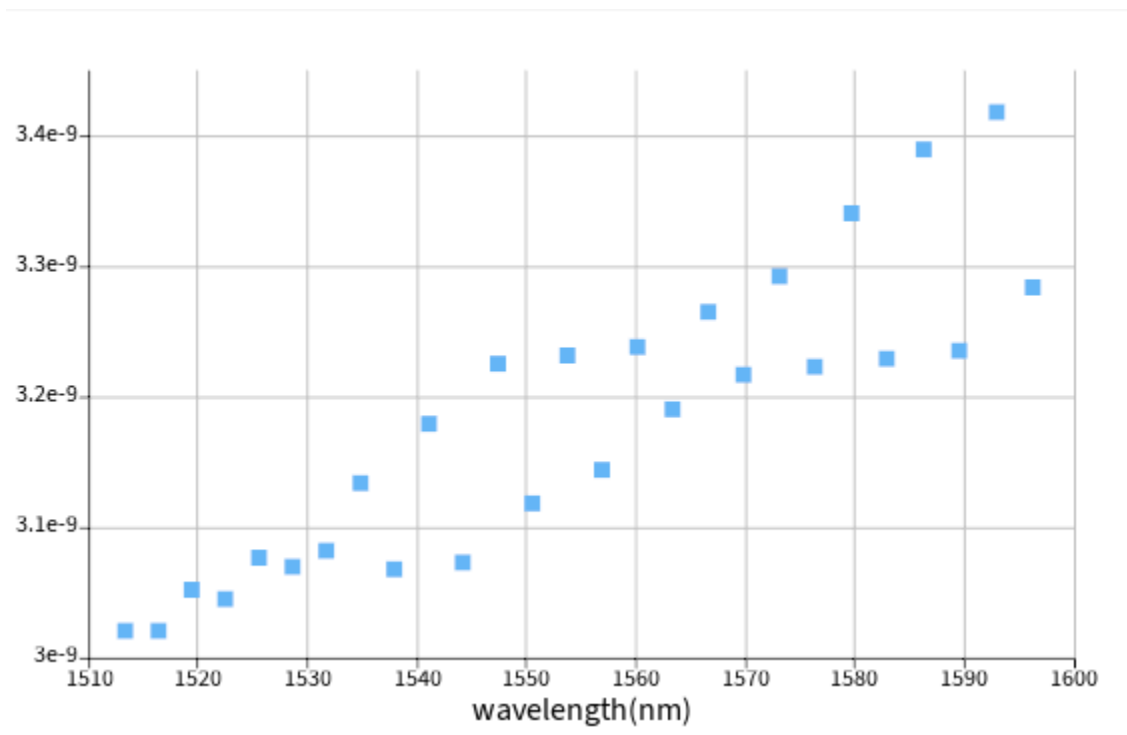
Spectrum of MZI delta L 160



FSR delta L 160 expected FSR = 3.57515e-09

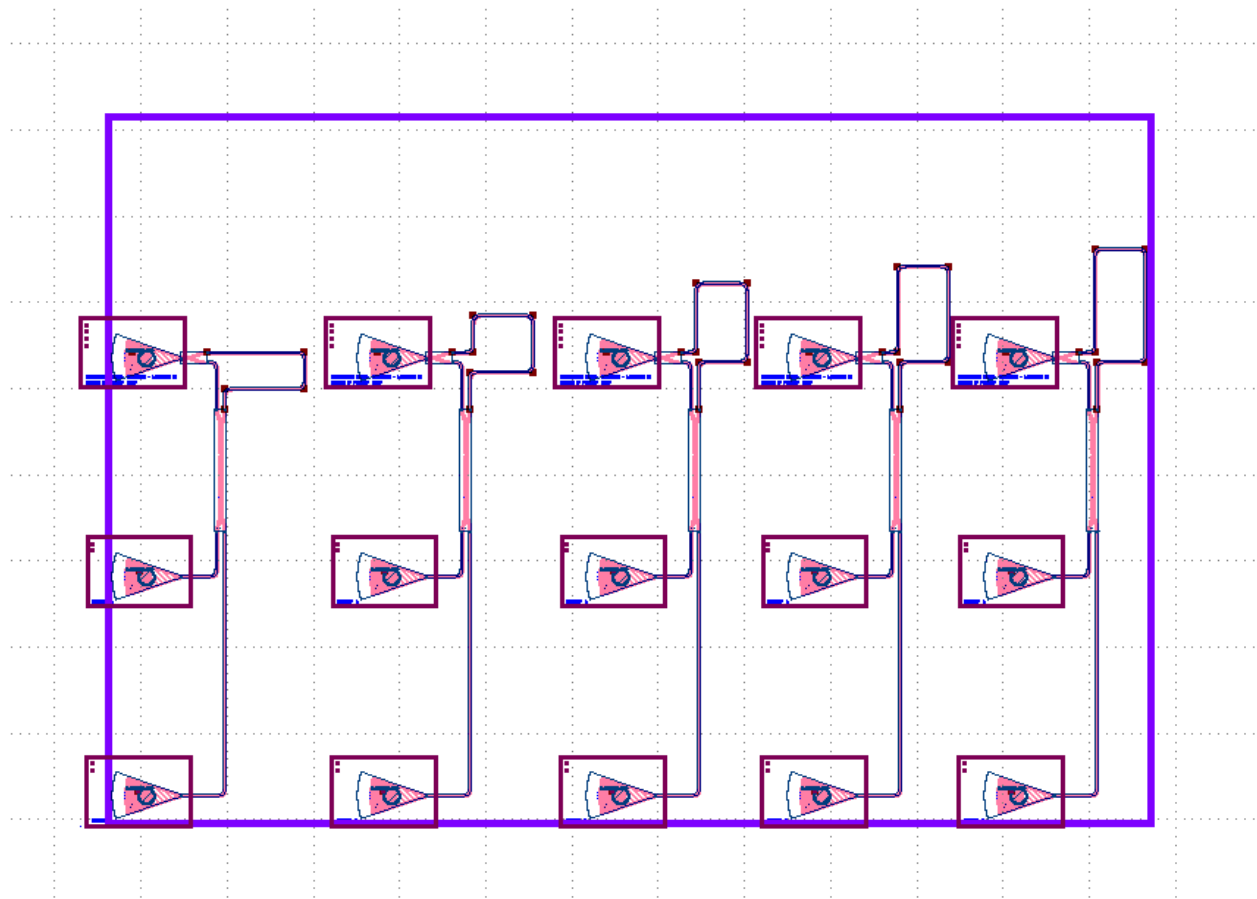


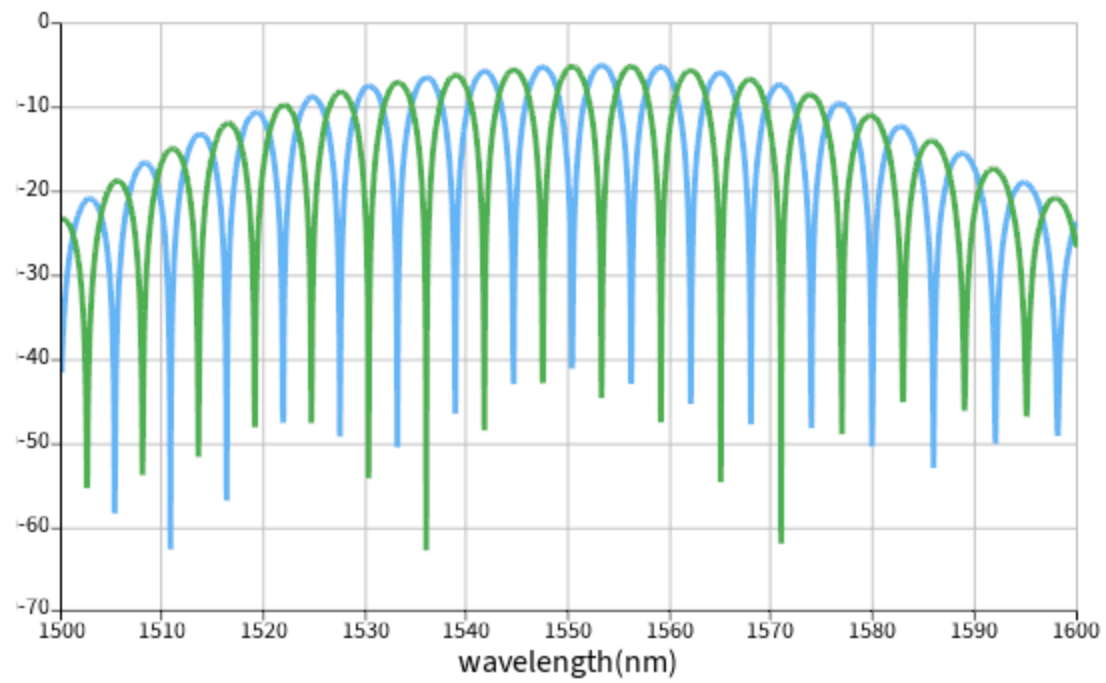
Spectrum of MZI delta L 180



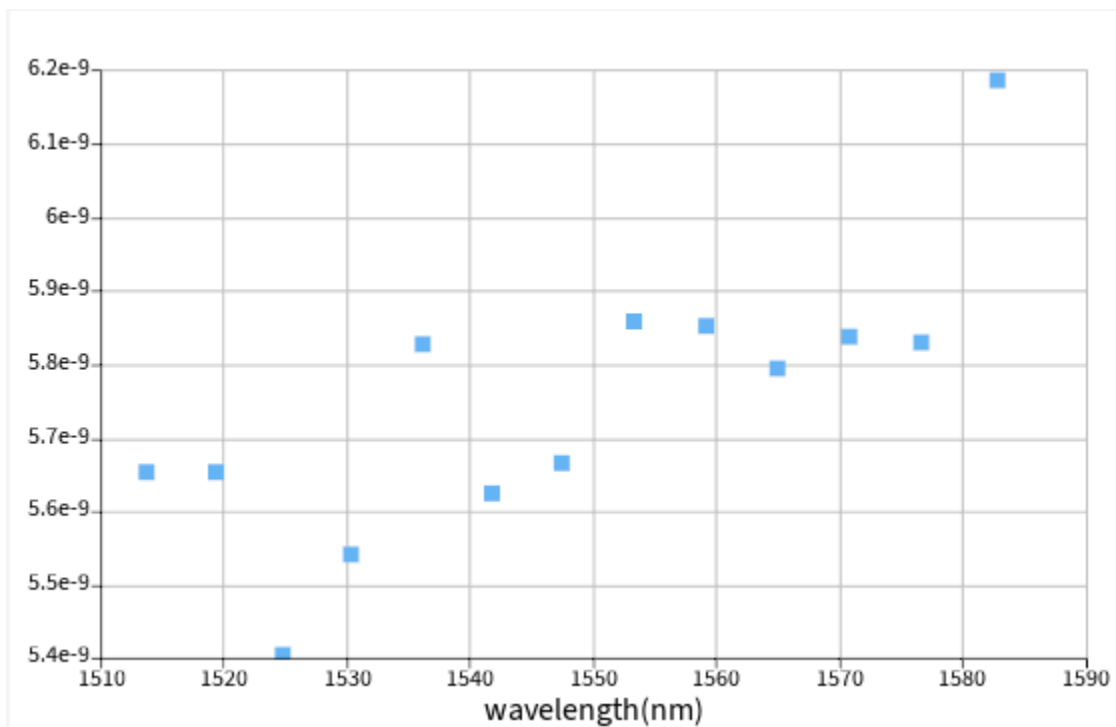
FSR delta L 180 expected FSR = 3.17791e-09

Layout and simulations in Lumerical interconnect for Layout:

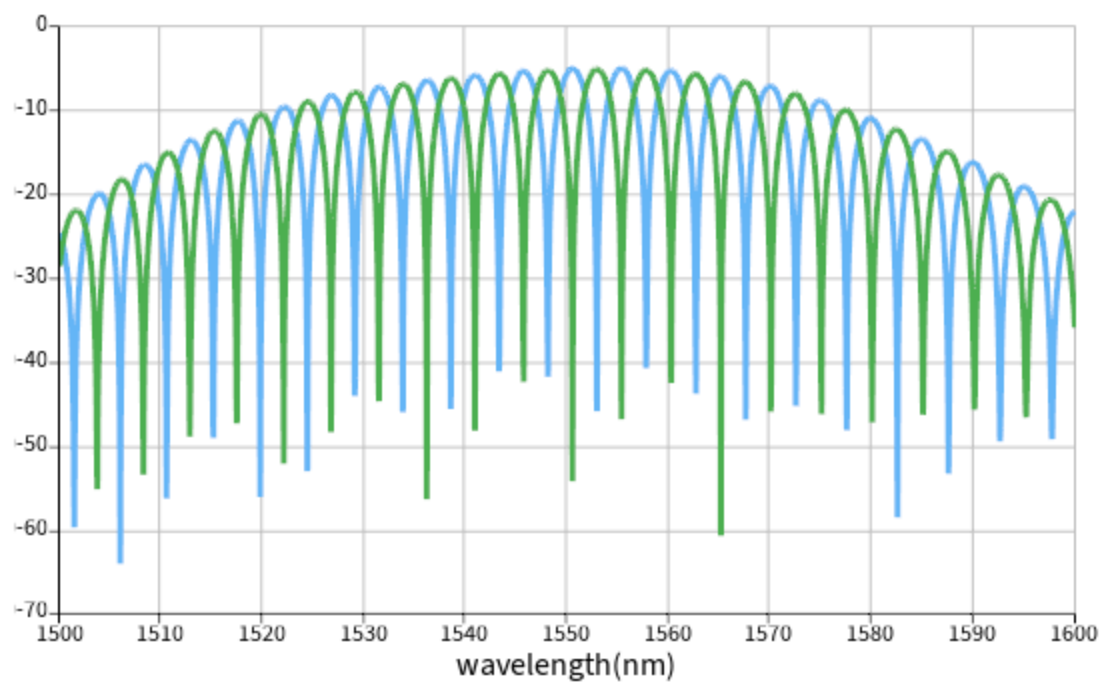




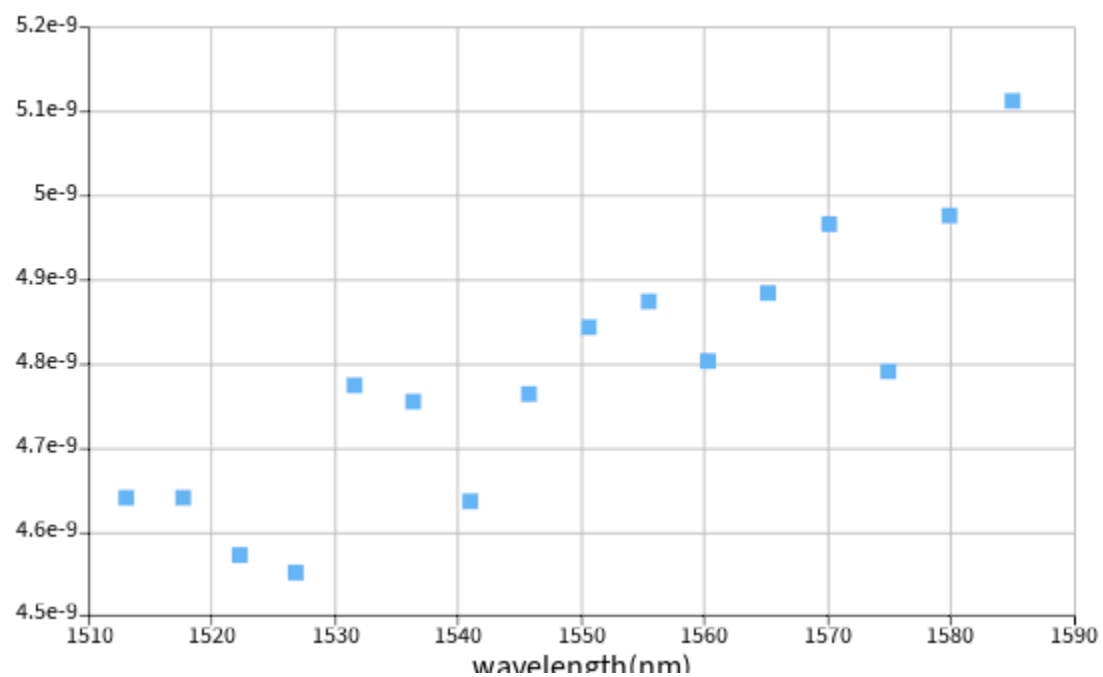
Simulation gain delta I 100



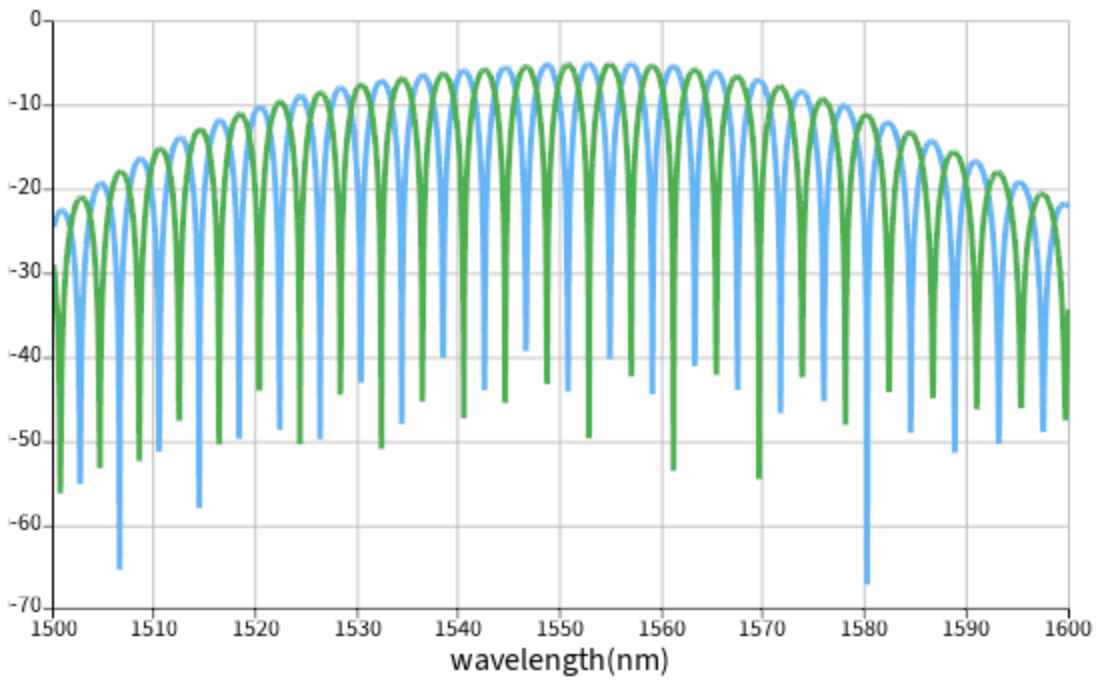
FSR simulation delta I 100



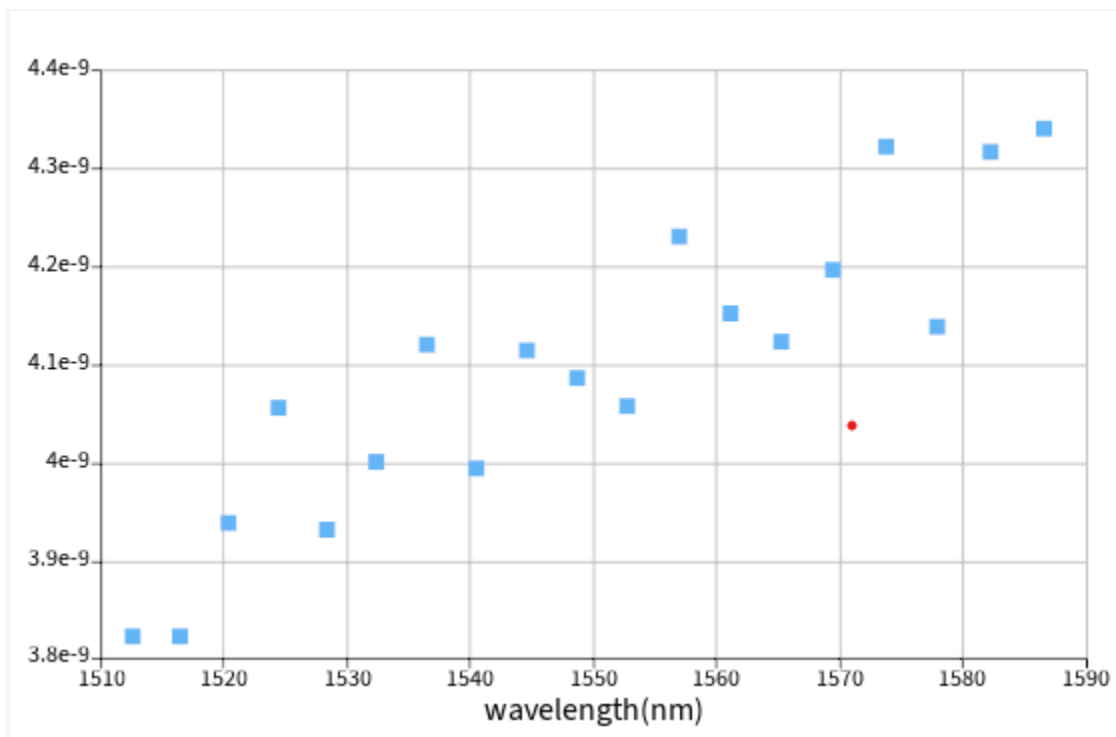
Simulation gain delta L 120



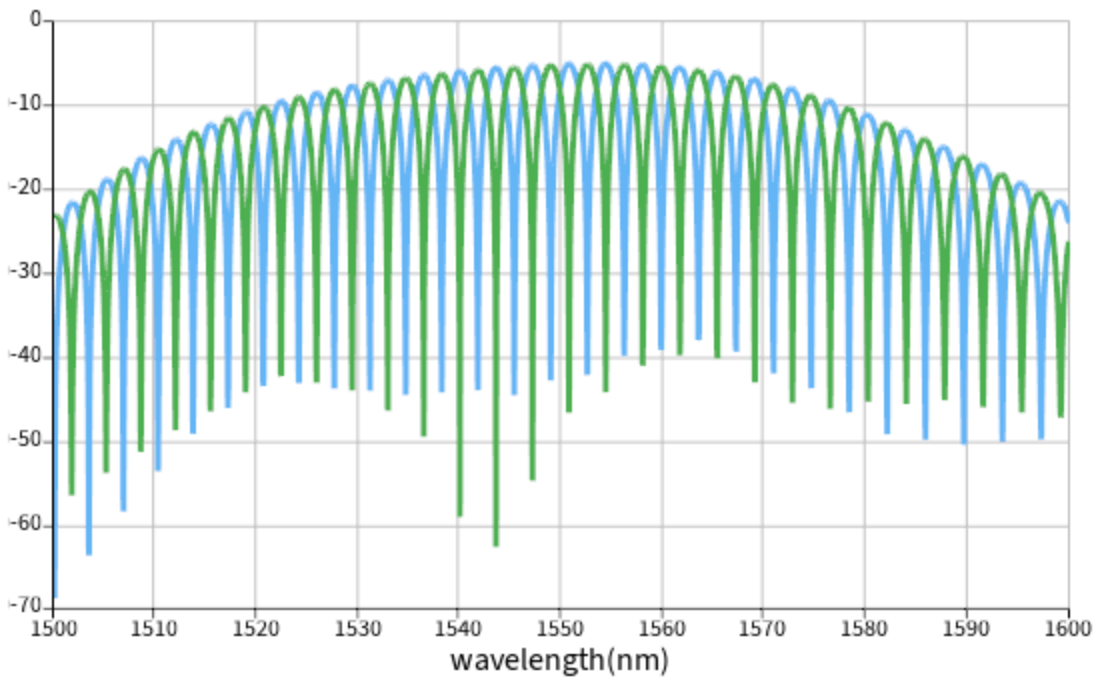
FSR simulation delta L 120



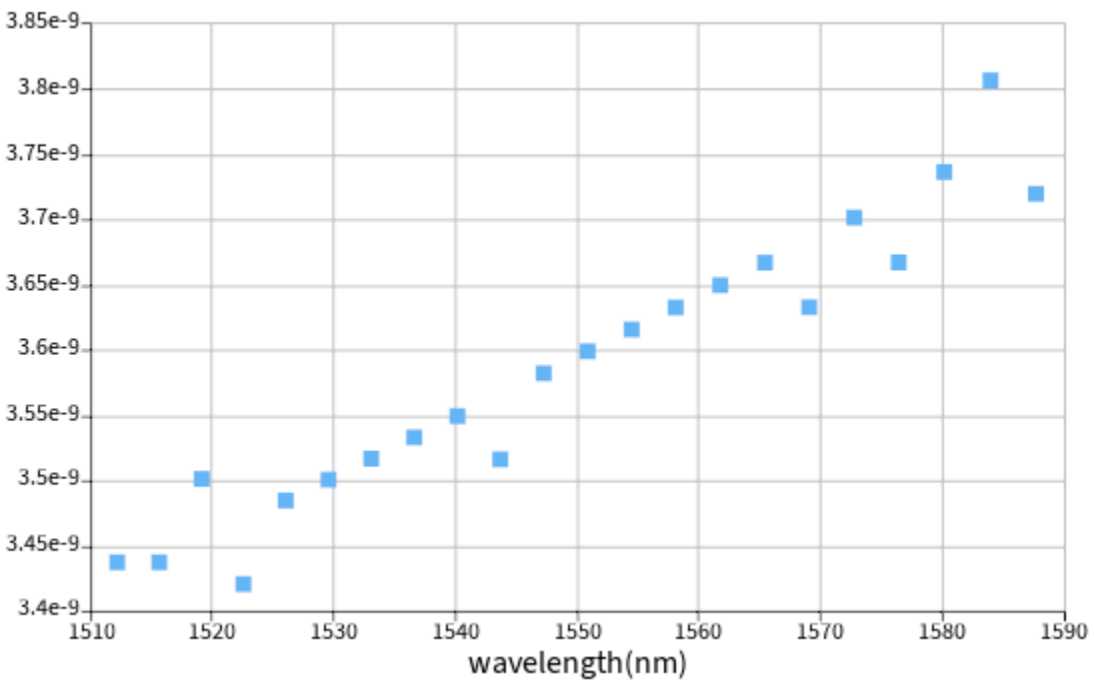
Simulation gain delta I 140



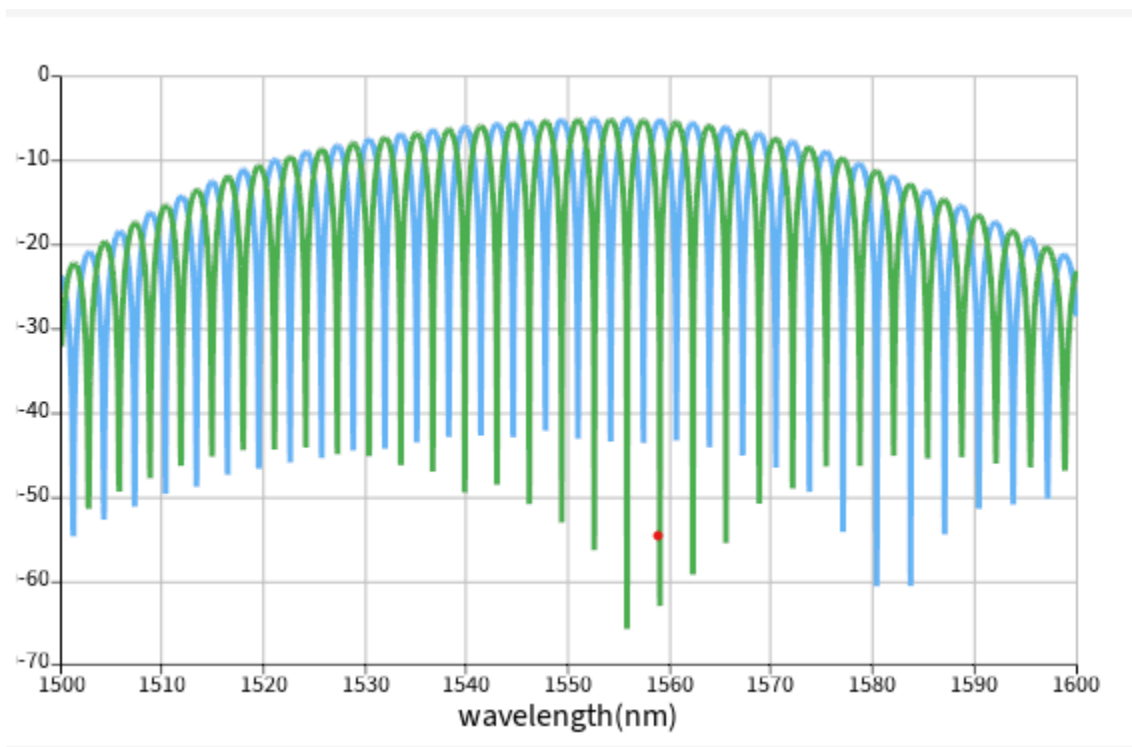
FSR simulation delta I 140



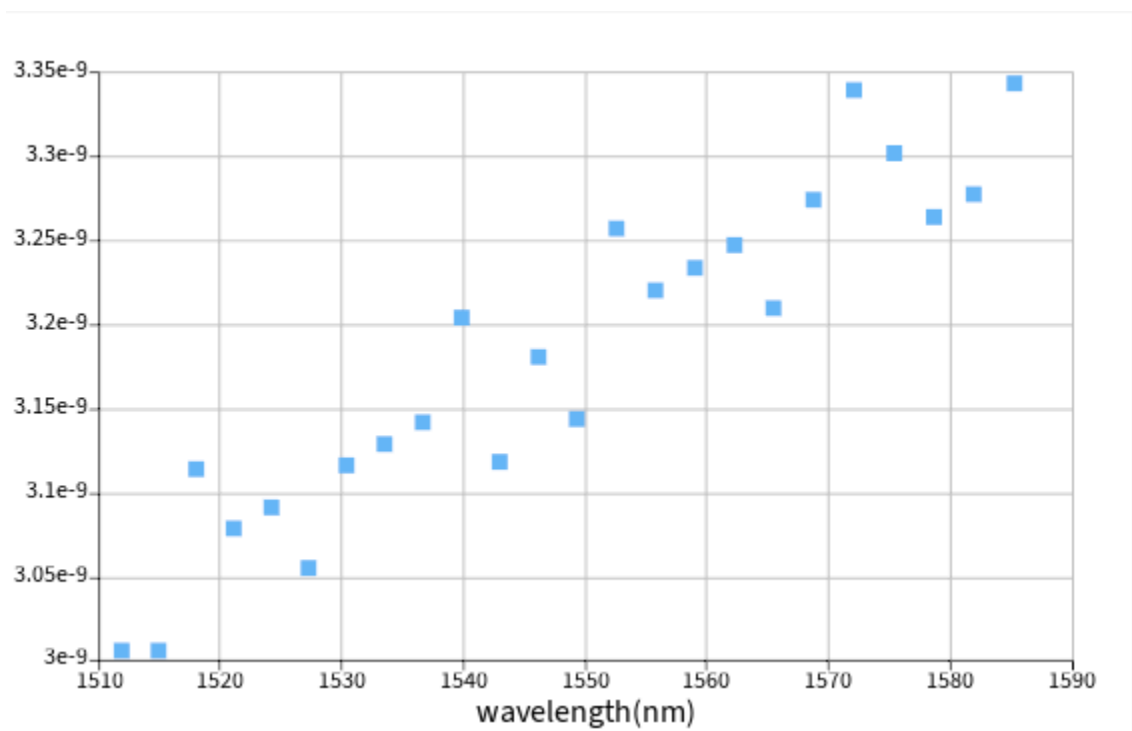
Simulation gain delta I 160



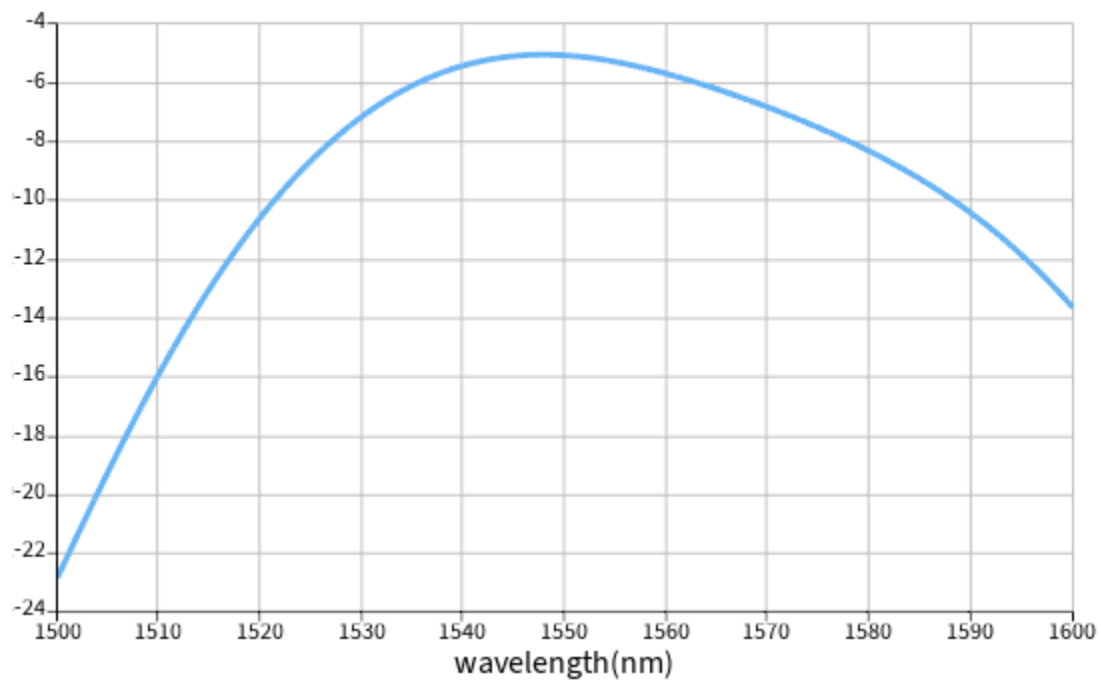
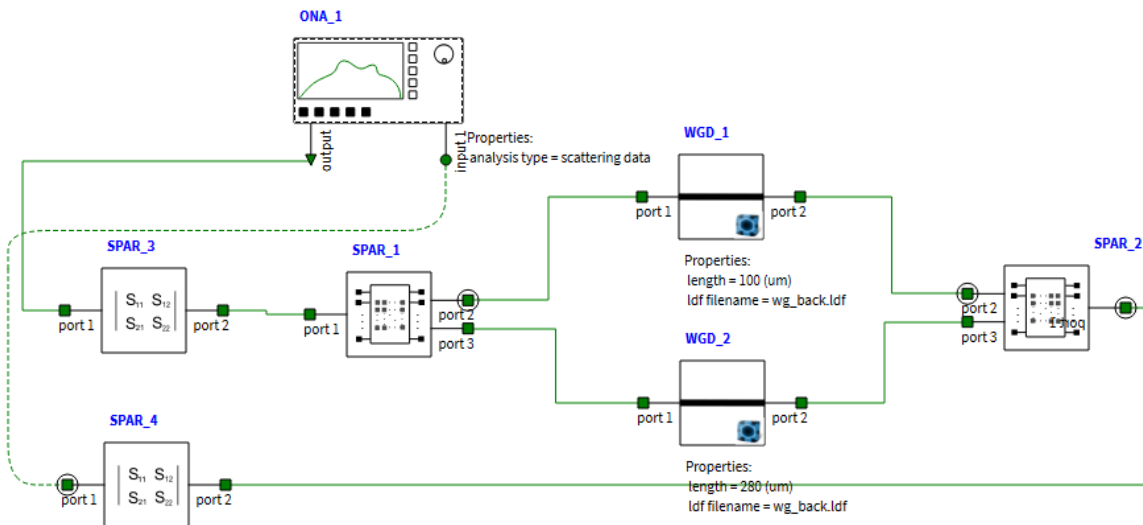
FSR simulation delta I 160



Transmission spectrum layout sim delta L = 180



FSR sim delta L = 180



Insertion loss grating coupler